Guide 2. APT Project Development Capstone Course

Personal Information

Below is a table where you must complete the requested information.

Student Name	National ID (RUT)	Degree Program	Campus
Lucas Joaquin Carvajal Pincheira	**21.146.055-5**	Computer Science & Software Engineering (Ingeniería en Informática)	Viña del Mar
Osvaldo Antonio Reyes Wiegand	**19.336.327-k**	Computer Science & Software Engineering (Ingeniería en Informática)	Viña del Mar

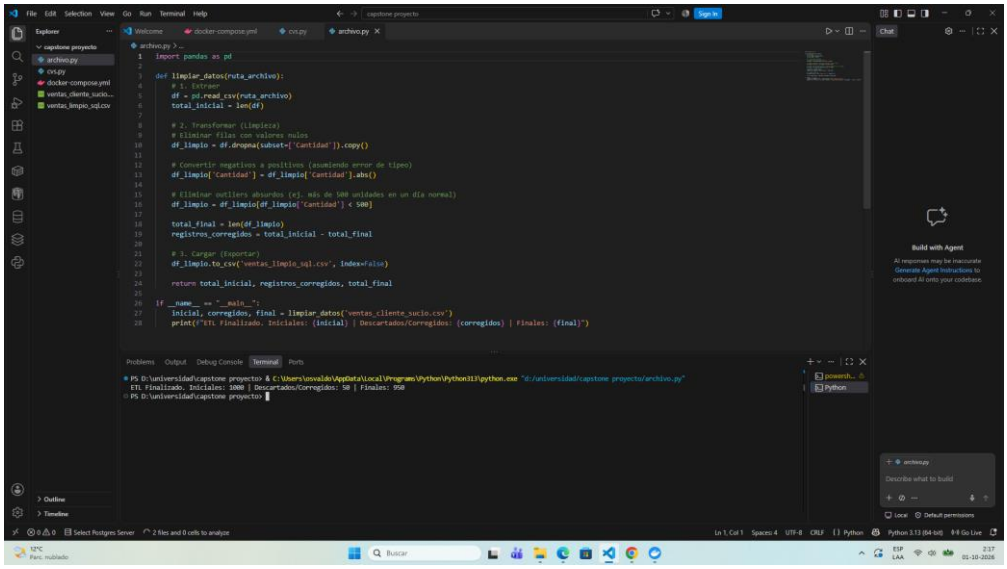
1. APT Project Progress Summary

Below you will find various fields that you must complete with the requested information.

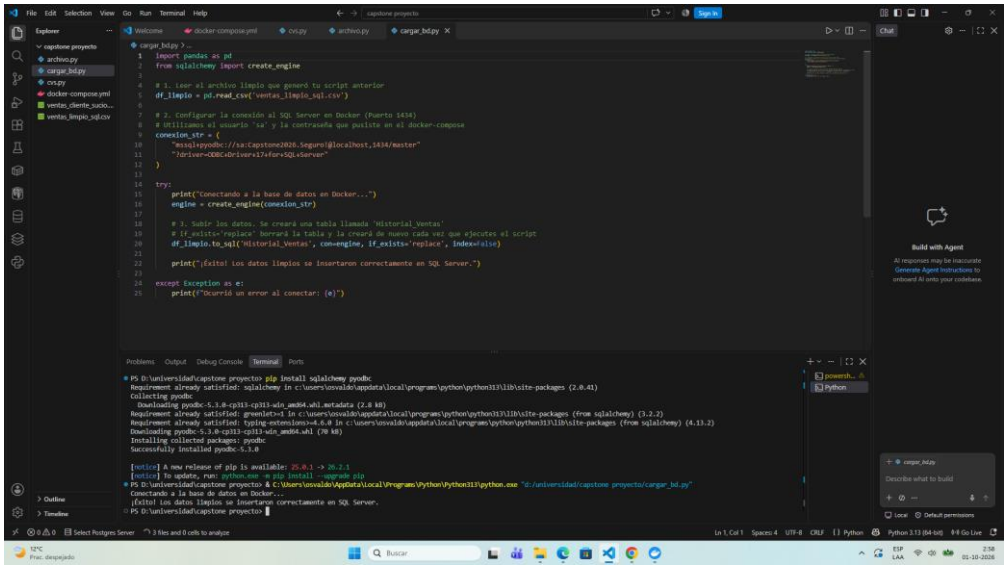
APT Project Progress Summary	During this stage, progress focused primarily on consolidating the data infrastructure and analytical baseline of the Predictive Demand and Inventory Optimization System for the wholesale apparel and footwear distributor in the Valparaíso Region (Quilpué / Viña del Mar). An automated Extraction, Transformation, and Loading (ETL) pipeline was successfully implemented in Python using Pandas, filtering out null values and correcting inconsistencies. Additionally, a containerized Microsoft SQL Server 2022 relational database engine was deployed via Docker Compose on port 1434 with persistent

	storage volumes, where the cleaned data was fully loaded using SQLAlchemy. Finally, the initial Machine Learning predictive model (Random Forest Regressor) was trained, achieving a baseline Mean Absolute Error (MAE) of 24.11 units, and the layout of the executive web dashboard in Streamlit was developed for early inventory alert visualization by the client.
Objectives	1. Develop a predictive software system based on Machine Learning to optimize inventory management and estimate future product demand in a textile and footwear wholesale company. 2. Implement an automated data ingestion, cleaning, and normalization pipeline (ETL) in Python to ensure the integrity of sales records. 3. Deploy and configure the relational/dimensional database in Microsoft SQL Server 2022 using Docker containers to guarantee persistence and centralization of transactional information. 4. Train and evaluate Machine Learning models (Random Forest / Time Series) using formal quantitative metrics (MAE/MAPE) to project seasonal purchasing volume. 5. Build an interactive web interface in Streamlit that allows the wholesale company user to visualize sales reports, demand forecasts, and preventive stockout alerts.
Methodology	The project continued to be developed under the Agile Scrum framework, adapted to the academic context through iterative and incremental deliveries oriented toward data architecture and applied artificial intelligence. The agile ceremonies and guidelines defined by the team were maintained: ● **Sprint Planning and Daily Meetings: Planning sessions were held at the beginning of each cycle along with remote synchronization meetings (2 to 3 times per week) to coordinate workloads across technical roles (Osvaldo Reyes in database architecture, ETL, and Machine Learning algorithms; Lucas Carvajal in requirements engineering, documentation quality assurance, test design, and KPI calculation).** ● **Sprint Review and Retrospective: At the end of each cycle, functional technical increments were reviewed (ETL scripts, Docker orchestration, SQL Server migration scripts, and AI training console), analyzing facilitators and blockers (such as unstructured historical client data and credential management) and agreeing on versioning and security improvements.** ● Technical work relied on Visual Studio Code, Python (Pandas, Scikit-Learn, SQLAlchemy, PyODBC), Microsoft SQL Server 2022 containerized on Docker Desktop, and Streamlit for the presentation layer. Unit tests, integration tests, and quantitative metrics were applied to back software quality.
Progress Evidence	The presented evidence corresponds to the deliverables committed in the formal Phase 1 document (1.5) and the software quality assurance guidelines: 1. **1. ETL Pipeline and Persistence Model (Milestone 2 / Progress 1): Execution of the Python script (archivo.py) to clean the client's database records and export them to ventas_limpio_sql.csv (950 records with 100% integrity), followed by**

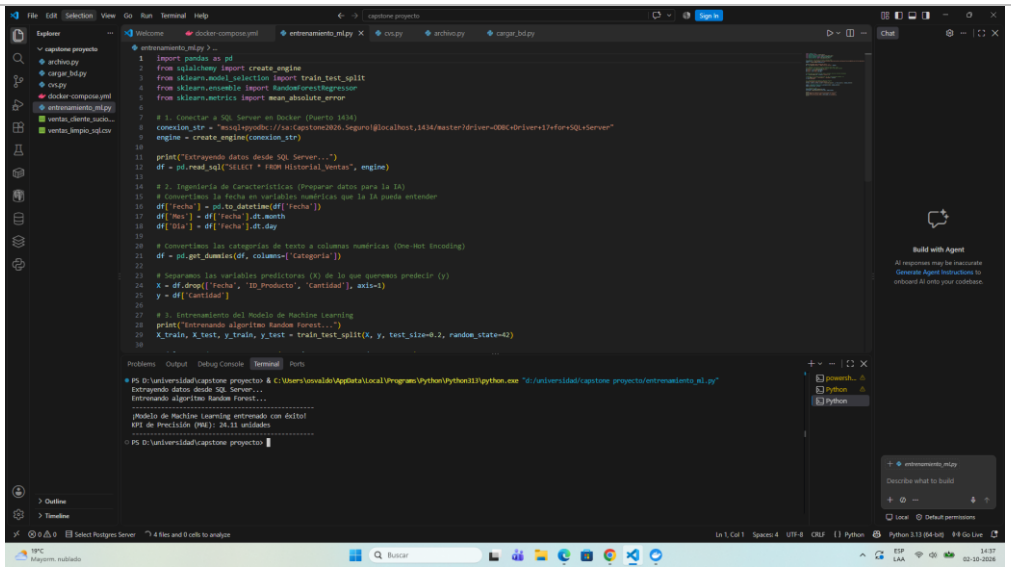
the bulk migration script (cargar_bd.py) connected to SQL Server.



2. **2. Containerized Infrastructure: Orchestration file docker-compose.yml running the sqlserver service (image mssql/server:2022-latest, container bd_capstone_sql on port 1434 with persistent volume sqlserver_data).**



3. **3. Artificial Intelligence Predictive Engine (Milestone 3): Machine Learning training source code (Random Forest in entrenamiento_ml.py) executed with an 80/20 train/test split, yielding a console-measured Mean Absolute Error (MAE) of 24.11 units per garment as a seasonal inference baseline.**



4. **4. Unit and Integration Test Cases: Formal execution of tests PU-01 (dropna), PU-02 (abs), PI-01 (ETL + CSV), and PI-02 (Python + Docker SQL Server) with Approved status.**

ID	Unidad probada	Entrada	Resultado esperado	Estado	Resultado obtenido
PU-01	dropna() (Manejo de nulos)	Fila con Cantidad = NaN	Eliminación de la fila	Aprobado	Fila descartada del dataset final
PU-02	abs() (Manejo de negativos)	Fila con Cantidad = -15	Conversión a 15	Aprobado	Valor normalizado a positivo

5. **5. Technical Metrics and KPIs: Formal registration of KPI-01 (Ingestion Quality: 100% of data processed with no anomalous values) and KPI-02 (Predictive Model Precision: baseline MAE of 24.11 units).**

7.5 Indicadores Clave de Rendimiento (KPIs)

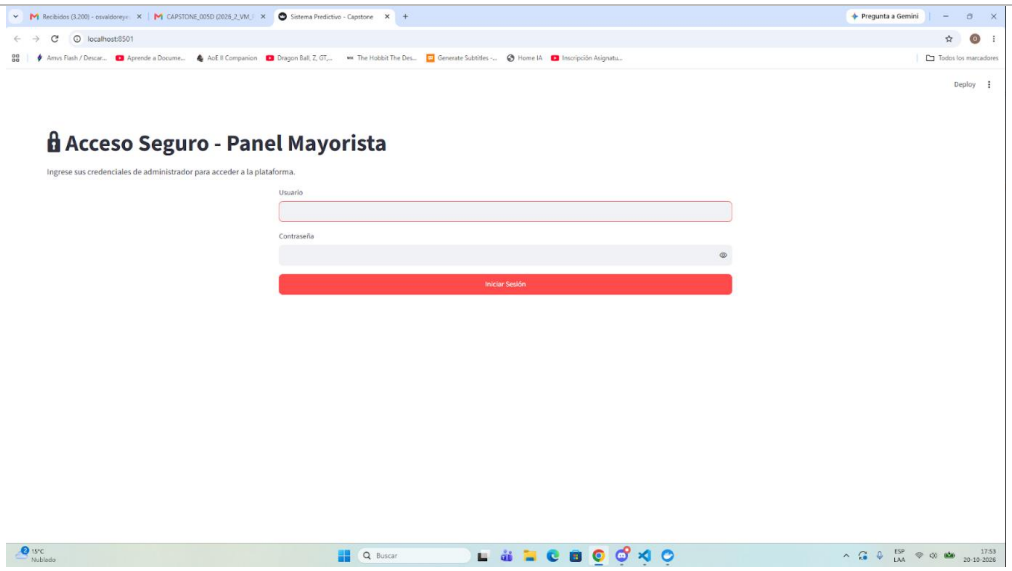
KPI-01: Calidad de Ingesta y Depuración de Datos

- **Objetivo:** Garantizar la integridad estructural y matemática de la serie histórica antes de suministrar los datos a los algoritmos predictivos.
- **Procedimiento:** Ejecución del pipeline ETL (`archivo.py`) sobre el conjunto de datos `ventas_cliente_sucio.csv` (1.000 transacciones).
- **Resultado Esperado:** 100% de registros nulos y valores negativos tratados o depurados, asegurando cero fallas por inconsistencia de datos.
- **Resultado Obtenido:** Sobre un total de 1.000 registros evaluados, el script identificó y descartó 50 filas con valores nulos (5%) y normalizó 49 registros con cantidades negativas. Se generó un conjunto final consolidado de 950 registros íntegros en `ventas_limpio_sql.csv`.

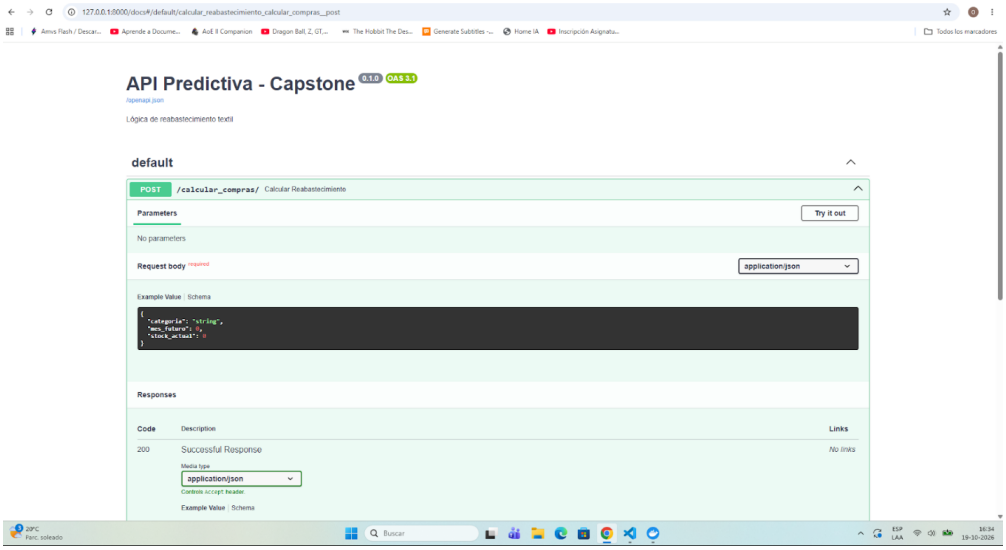
6. Backlog and Dashboard: Local deployment in Streamlit (app.py) featuring graphical visualization of historical sales by category and an inventory control panel.

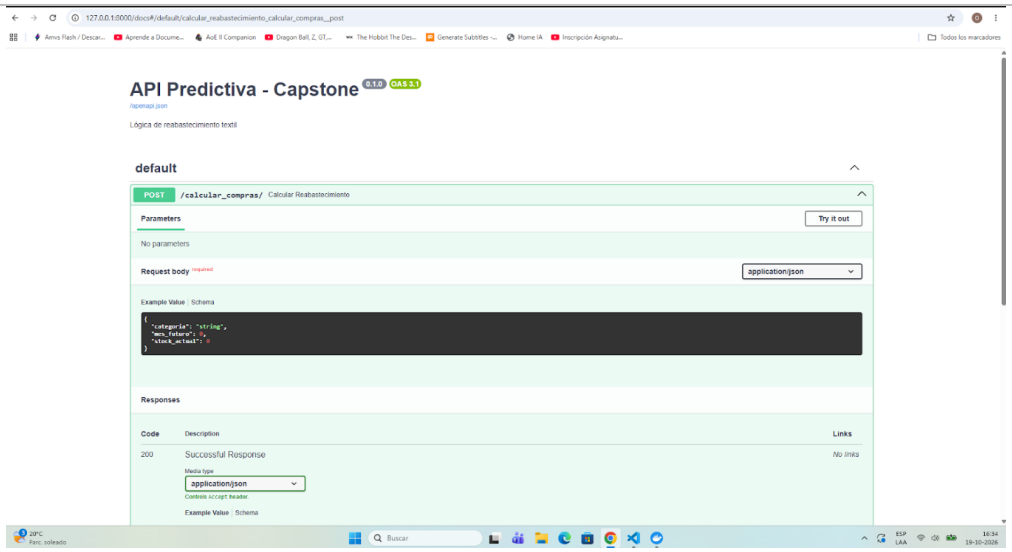


7. Login Security: Implementation of a secure login interface (Secure Access - Wholesale Panel) with credential authentication for users and administrators.

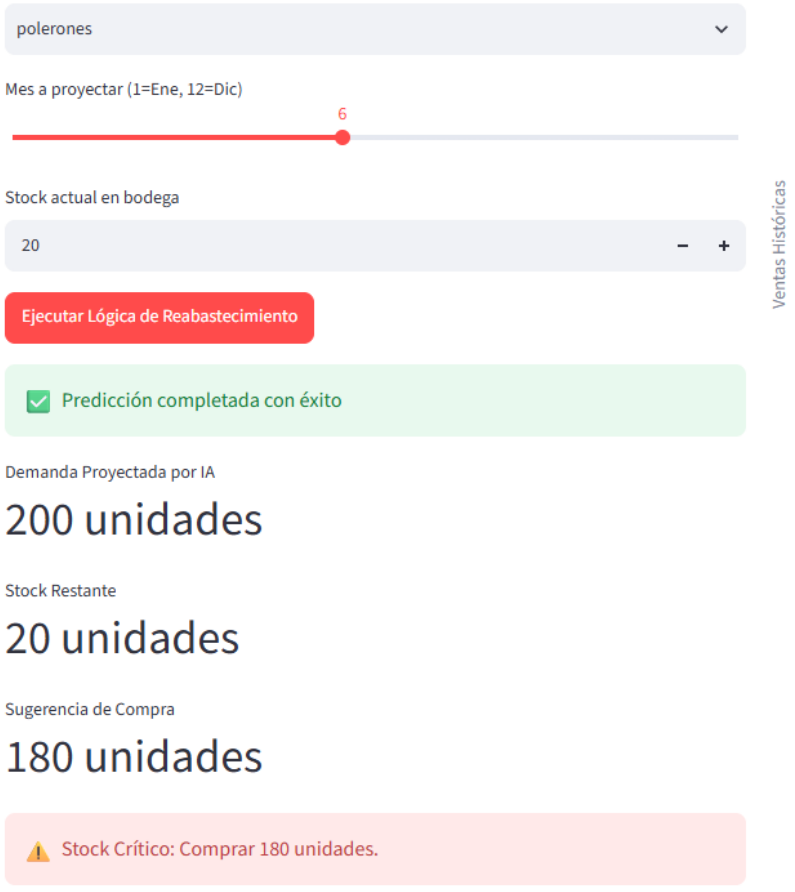


8. Predictive API: Construction and documentation in FastAPI (Swagger UI) of the /calcular_compras/ endpoint for communication between the AI model and the backend.





9. Interactive Predictive System: Interactive frontend module for warehouse stock input, monthly demand projection, and automatic alert calculation for critical stock.



2. Work Plan Monitoring

Carefully examine your work plan, focusing especially on the progress status and adjustments columns.

APT Project Work Plan

Competency or Competency Units	Activities	Resources	Activity Duration	Responsible	Observations / Remarks	Progress Status	Adjustments
Requirements Engineering	Phase 1: Requirements gathering and definition of inventory variables	Virtual/in-person meeting with client, Text documents	1 week	Lucas Carvajal	*Clarity on wholesale purchasing challenges; the obstacle was the lack of formal structure in the client's historical records*	*Completed*	*No adjustments made*
Agile Management (Scrum)	Phase 1: Role definition, Product Vision, and prioritized Backlog	Scrum templates, Visual Studio Code	1 week	**Lucas Carvajal and Osvaldo Reyes**	*Facilitated an equitable division between data/AI engineering and documentation QA; the obstacle was coordinating delivery schedules*	*Completed*	*No adjustments made*
Data Architecture and Management	Phase 1: Architecture definition, persistence strategy, and Docker setup	Docker Desktop, docker-compose.yml	2 weeks	Osvaldo Reyes	*Containerization guaranteed immediate portability; the obstacle was the local port 1433 conflict*	*Completed*	*No adjustments made*
Data Engineering (ETL)	Phase 2 - Sprint 1: Client data ingestion and ETL cleaning pipeline	Python, Pandas, Visual Studio Code	2 weeks	Osvaldo Reyes	*Using Pandas enabled vectorized cleaning; the obstacle was normalizing unstructured records across 4 product categories*	*Completed*	*Processing and cleaning of 1,000 client transactional rows*
Relational Databases	Phase 2 - Sprint 1: Migration and bulk insertion into Docker SQL Server	SQLAlchemy, PyODBC, Docker Container	1 week	Osvaldo Reyes	*Robust connection without record loss; the obstacle was ODBC driver compatibility in the operating system*	*Completed*	*No adjustments made*
Artificial Intelligence	Phase 2 - Sprint 2: Baseline predictive	Python, Scikit-Learn (Random	2 weeks	Osvaldo Reyes	*Fast computation of the tree ensemble; the challenge was*	*Completed*	*No adjustments made*

	model training and MAE calculation	Forest), VS Code			*balancing the 80/20 validation split*		
QA / Testing and Quality	Phase 2: Unit tests, integration tests, and KPI formalization	Testing scripts, VS Code Terminal	2 weeks	Lucas Carvajal	*Tests PU-01 to PI-02 approved; the challenge was capturing exact execution times and metrics in console*	*Completed*	*KPI-01 and KPI-02 technical sheets were formalized*
Frontend / UI Development	Phase 2 - Sprint 3: Dashboard layout and alert module	Streamlit, Python, Visual Studio Code	2 weeks	Osvaldo Reyes and Lucas Carvajal	*Streamlit enables rapid development in Python; the challenge was connecting the replenishment logic with the charts*	*Completed*	*Visual components developed and integrated*
Software Security	Phase 2: Environment variable management and network isolation	.env, .env.example, .gitignore files	1 week	Lucas Carvajal	*Isolates credentials from the repository; the challenge was removing plaintext passwords from legacy scripts*	*Completed*	*Passwords decoupled into .env file*
Technical Documentation	Phase 2: Consolidation of Phase 2 report, technical manual, and README	Word, Markdown, Architecture Diagrams	1 week	Lucas Carvajal	*Complete backing of evidence; the challenge is the time required to diagram the end-to-end architecture*	*In Progress*	*No adjustments made*

3. Adjustments Based on Monitoring

Delve into the observations of your work plan. Analyze the planned activities and indicate which aspects facilitated or hindered the execution of the plan. Explain how you addressed and/or will address the obstacles. Finally, state the adjustments made to the work plan based on this analysis.

Factors that have facilitated and/or hindered the development of the work plan:

Facilitating Factors: The strict division of roles and disciplinary responsibilities between team members (Osvaldo Reyes in technical architecture, script development, and algorithmic modeling; and Lucas Carvajal in requirements management, test case design, and documentation QA). Furthermore, using modern tools like Docker Desktop allowed rapid deployment of a standardized and reproducible SQL Server 2022 database engine, reducing local software configuration times.

Obstructing Factors: The integration of Microsoft SQL Server via Docker Desktop represented an initial technical challenge in the project's data architecture. Initially, the transition hindered progress due to network conflicts with the default port (1433) and compatibility issues with ODBC drivers in the local environment. However, this obstacle was overcome by reconfiguring the Docker Compose

orchestration to external port 1434 and adjusting the connection strings using SQLAlchemy and PyODBC in Python, ultimately guaranteeing data persistence, isolation, and portability.

Adjusted or Removed Activities:

Database Port Reconfiguration: The network configuration in the docker-compose.yml file was adjusted to bind the service through port 1434:1433, adapting the Python scripts' connection string to localhost,1434 without altering container integrity.

No Activities Removed: All tasks associated with software engineering, data architecture, Machine Learning model development, and user interface development remain active within the project scope.

Activities Not Started or Delayed: According to the schedule established in Document 1.5 from Phase 1, we are ahead of schedule regarding the joint work of Phase 2. In-progress activities are minimal, the system is functional and requires only minor refinements, data ingestion has been completed, and final documentation is currently in progress.

Critical infrastructure activities, ETL cleaning pipelines, SQL Server loading, Machine Learning training, and unit/integration testing are 100% completed.

Tasks currently in progress include final refinements to the Streamlit web interface (app.py) for inventory alert deployment, drafting the technical manual, and consolidating the official report.

The activity that has not yet started corresponds to Phase 3: Final user stress testing, field operational validation, and the formal Capstone Portfolio defense before the evaluation committee.

Guía Estudiante - Desarrollo Proyecto APT
Fase 2